
SUPERVISOR PLANNING MEETING

CHRONO-Fair

Time-Resolved Counterfactual Fairness Monitoring
for Clinical Machine Learning

Extends our prior VFR-Audit three-axis static auditing framework

Target venue: Q1 digital-health / health-informatics journal

Anonymous draft / in-progress research / not finalised

Where clinical ML stands today

🏏 Clinical ML supports triage, risk scoring, and length-of-stay prediction across many sites [1, 2].

⚠️ Documented post-release subgroup failures across radiology, sepsis, kidney function, and care-allocation models [3, 4, 5, 6].

🏛️ Regulatory expectations have shifted toward continuous lifecycle monitoring [7, 8] with reporting standards [9, 10, 11, 12].

✂️ Auditing toolkits in routine use return a thresholded pass-or-fail verdict on a fixed snapshot [13, 14, 15, 16].

That verdict is the object hospitals act on. The snapshot tells us nothing about its stability over time.

Three audit-reliability axes on real Texas-100X (925 128 records) [17]:

- Axis 1: resampling fluctuation of the fairness verdict.
- ✂ Axis 2: audit-size sensitivity of the fairness verdict.
- 🏠 Axis 3: cross-hospital-site agreement of the fairness verdict.

Standard vs Fair monitored-model setup (cited from prior work):

Attribute	Std DI	Fair DI	Verdict
Race	0.654	0.884	fail → pass
Sex	0.762	0.933	fail → pass
Ethnicity	0.832	0.962	pass → pass
Age Group	0.291	0.801	fail → pass

Std Acc 0.8777 → Fair Acc 0.8248 (reduction of 5.28 pp). Predecessor, not a CHRONO-Fair contribution.

The remaining gap

© **Real Texas-100X anchor finding (925 128 records) [17].** The worst-group true-positive-rate verdict for race reads “fair” at **0.824**. The same verdict flips to “unfair” in **4 of 20** hospital-network resamples (Verdict Flip Rate **0.20**). A batch audit cannot tell when, or whether, that verdict reversal recurs as new patients arrive.

Gap 1, time axis.

Batch verdicts give no times-tamp [13, 14, 15].

Gap 3, diagnostic interpretation.

Without triage, a team cannot tell whether to fix data or collect more samples [21, 22].

Gap 2, peeking penalty.

Repeated inspection inflates false-alarm rates without a sequential guarantee [18, 19, 20].

Gap 4, regression output.

LOS outputs lack a flip-rate or verdict-stability extension [23].

Closest related work, named and positioned

Tool / work	Mode	Output	Closest gap
Aequitas [13]	Batch	Per-attribute parity-ratio report	No streaming alarm
Fairlearn [14]	Batch	Group fairness + reduction mitigator	No verdict-flip monitoring
AIF360 [15]	Batch	20 metrics library	Snapshot only
Microsoft RAI [16]	Notebook	Fairlearn + interpret + CF	Developer review, not governance
Davis et al. [2]	Quarterly	Subgroup trend, 11 yr VA	No per-cell sequential alarm
Prediction sensitivity [22]	Per batch	CF-flip rate scalar	No alarm rule
VFR-Audit (our prior) [17]	Static	Verdict Flip Rate scalar	No time axis

Anytime-valid machinery exists [18, 19, 20]. It has not been applied to counterfactual fairness verdicts at intersectional granularity.

Our planned extension, in one sentence

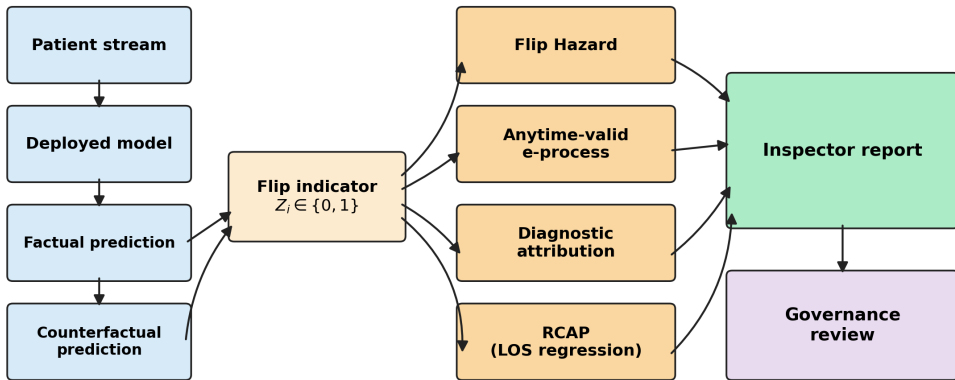
CHRONO-Fair contributes a post-release, time-resolved monitoring layer for thresholded fairness verdicts using a per-cell anytime-valid e-process on counterfactual flip indicators.

 **Monitor, not mitigator**  **Begins after Std → Fair**

 **Agnostic to mitigation method**  **Research prototype, not medical device**

Planned architecture

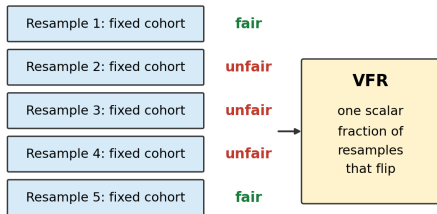
CHRONO-Fair: patient stream to Inspector report



Modules: Flip Hazard [24]; per-cell shrunk-GROW e-process [18, 19]; aleatoric/epistemic decomposition [21]; RCAP Wasserstein-1 rank shift [23]

How CHRONO-Fair extends VFR-Audit

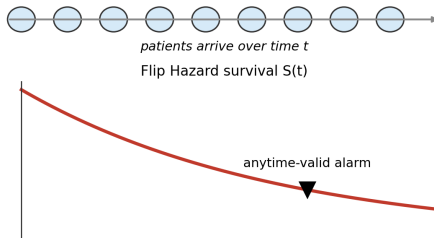
VFR-Audit (prior work): static batch audit



Computed once, before deployment.

No time axis. No alarm. No diagnostic triage.

CHRONO-Fair: streaming patient-arrival monitoring



Scalar VFR is recovered as $1 - S$ at the audit horizon.

Adds: when (time), where (cell), how to triage.

Scalar VFR is recovered as $1 - S$ at the audit horizon. CHRONO-Fair adds the time axis, the anytime-valid alarm rule, and the diagnostic triage signal.

Equation / construct	Attribution
Flip indicator Z_i	Our definition ; same idea as prediction sensitivity [22]
Flip Hazard $\lambda_a(t)$	Standard hazard form (Kaplan-Meier [24]; Nelson-Aalen 1972); our naming and application
RMFT integral	Standard Restricted Mean Survival Time; adopted
e-process wealth E_t	Waudby-Smith and Ramdas [29]; Ramdas et al. [20]; adopted
Shrunk GROW bet λ_s	Our finite-sample shrinkage of the GROW rule of [29]
Ville's inequality (Thm. 1)	Ville 1939 [18]; standard application
Ensemble flip probability $\bar{p}^{\text{flip}}$	Our definition
Aleatoric/epistemic entropy split	Depeweg et al. [30]; Hullermeier and Waegeman [21]; adopted
RCAP rank positions u_i	Standard nonparametric group-conditional CDF; our naming
RCAP $= W_1(\dots)$	Wasserstein-1 from optimal transport; adopted

The novelty is not a new survival estimator or a new e-process. The contribution is the coordinated integration of these instruments around the thresholded fairness verdict.

How each planned module closes one identified gap

Gap 1, time axis



Flip Hazard. Per-arrival survival curve instead of a single scalar [24].

Gap 2, peeking penalty



Per-cell anytime-valid e-process. Type-I control at every inspection step [18, 19, 20].

Gap 3, diagnostic interpretation



Aleatoric/epistemic decomposition. Triage label (data, model, mixed) per alarmed cell [21].

Gap 4, regression output



RCAP, Wasserstein-1 on ranks. Counterfactual rank shift for LOS regression [23].

- ✓ **Per-cell anytime-valid Type-I control:** YES, under Ville inequality [18, 19].
- ✗ **Time-uniform online FDR:** NO. Step-wise BH is applied at inspection time only [25, 26].
- ✗ **Causal root cause:** NO. Diagnostic attribution is aleatoric/epistemic triage [21].
- ✗ **SCM-strength counterfactual:** NO. Feature-shift only; may under-flag indirect effects [27, 28].
- ✗ **Prospective clinical deployment:** NO. Controlled replay and simulation only.

Method	Detected	Mean delay (patients)
CHRONO-Fair (per-cell e-process)	50 / 50	837 (median 828)
ADWIN ($k = 1$)	21 / 50	3 878
ADWIN ($k = 25$)	21 / 50	3 927
Davis-style ADWIN on fairness gap	21 / 50	3 904
Periodic batch ($n = 500$)	24 / 50	499 (no peeking control)
Static VFR	0 / 50	—

10 000-patient stream, drift onset at 5 000, 50 Monte-Carlo runs. False-alarm rate at or below the nominal α for every tested α [19, 29].

Streaming evidence is controlled replay and simulation. Not prospective clinical validation.

Plan: how the gap will actually be closed

Phase 1] **Controlled-validation evidence.** Real Texas-100X batch + Texas-100X-calibrated streaming/replay + robustness stress tests.

Status: drafted, in revision for Q1 submission.

Phase 2] **Methodological strengthening.** Time-uniform online FDR [25, 26]; SCM counterfactual sensitivity [27, 28].

Status: planned for next revision.

Phase 3] **Prospective evaluation.** Site-partnered EHR-stream validation; governance-user evaluation.

Status: scoping, partner search.

Phase 4] **Artifact release.** Anonymised proof-of-concept prototype + configs + seeds + scripts.

Status: scheduled for camera-ready.

Asks for this meeting

1. Is the post-release monitoring framing the right contribution to lead with, vs a methods-only framing of the per-cell e-process?
2. Is the Std/Fair monitored-model table (cited from prior VFR-Audit study) acceptable as the predecessor without re-running it?
3. Which Q1 venue is best: Frontiers in Digital Health, npj Digital Medicine, JAMIA, JBI, or NEJM AI?
4. Should I add a second real cohort (MIMIC-IV / eICU) before submission, or submit single-dataset with a prospective-validation roadmap?
5. Defer artifact release to camera-ready, or supply a minimum-viable artifact preview at submission?
6. Timing: next-round submission in 6 to 8 weeks, or invest 2 to 3 months in Phase 2 strengthening first?

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